

Homework4

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1 Question1

MAP Estimation

Poisson(λ), with $S = \sum_{i=1}^n x_i$,

$$L(\lambda) \propto \lambda^S e^{-n\lambda}.$$

$\lambda \sim \Gamma(2, 1)$:

$$\pi(\lambda) = \lambda e^{-\lambda}, \quad \lambda > 0.$$

Also

$$\pi(\lambda \mid \mathbf{x}) \propto L(\lambda)\pi(\lambda) \propto \lambda^{S+1} e^{-(n+1)\lambda}.$$

$$\ell(\lambda) = (S+1) \ln \lambda - (n+1)\lambda + C.$$

$$\ell'(\lambda) = \frac{S+1}{\lambda} - (n+1) = 0$$

$$\ell(\lambda_{MAP}) = \frac{S+1}{n+1} = \frac{\sum_{i=1}^n x_i + 1}{n+1}$$

concavity

$$\ell''(\lambda) = -\frac{S+1}{\lambda^2} < 0 \quad \text{for all } \lambda > 0$$

$S+1 > 0$, so $\ell(\lambda)$ is strictly concave

$\hat{\lambda}_{MAP}$ is the unique global maximizer.

2 Question2

MLE

Likelihood: For i.i.d. $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$,

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \Rightarrow \ell(\lambda) = \ln L(\lambda) = \left(\sum_{i=1}^n x_i \right) \ln \lambda - n\lambda - \sum_{i=1}^n \ln(x_i!).$$

$$\ell'(\lambda) = \frac{\sum_{i=1}^n x_i}{\lambda} - n = 0 \Rightarrow \hat{\lambda}_{MLE} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

$\ell''(\lambda) = -\sum x_i \lambda^2 < 0$, confirming a maximum.
Meanwhile:

$$\ln p_X(x; \lambda) = x \ln \lambda - \lambda - \ln(x!), \quad \frac{\partial \ln p_X}{\partial \lambda} = \frac{x}{\lambda} - 1, \quad \frac{\partial^2 \ln p_X}{\partial \lambda^2} = -\frac{x}{\lambda^2}.$$

$$I_1(\lambda) = -E \left[\frac{\partial^2 \ln p_X}{\partial \lambda^2} \right] = \frac{E[X]}{\lambda^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

For the full sample:

$$I_n(\lambda) = n I_1(\lambda) = \frac{n}{\lambda}.$$

Asymptotic distribution of the MLE

Using MLE asymptotic theory,

$$\sqrt{n}(\hat{\lambda}_{MLE} - \lambda) dN(0, I_1(\lambda)^{-1}) = N(0, \lambda),$$

Approximately,

$$\hat{\lambda}_{MLE} \sim N\left(\lambda, \frac{\lambda}{n}\right).$$

Comparison with the CLT

Ordinary CLT

$$\sqrt{n}(\bar{X} - \lambda) dN(0, \text{Var}(X_i)) = N(0, \lambda),$$

$$\bar{X} \sim N\left(\lambda, \frac{\lambda}{n}\right).$$

$1/I_n(\lambda) = \lambda/n$ exactly matches the variance predicted by the CLT.

This is expected for the Poisson distribution because the MLE coincides with the sample mean.

Also the model achieves the Cramér–Rao lower bound.

3 Question3

Least Squares for the Linear Gaussian Model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \epsilon_i, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2), \quad i = 1, \dots, n.$$

With $\mathbf{Y} \in R^n$ and $\beta = (\beta_0, \dots, \beta_p)^T$:

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon, \quad \epsilon \sim N(\mathbf{0}, \sigma^2 I_n).$$

Given $X_1, \dots, X_p$, each $Y_i \mid \mathbf{X}_i \sim N(\mathbf{x}_i^T \beta, \sigma^2)$,
 So

$$p(Y_i \mid \mathbf{X}_i; \beta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - \mathbf{x}_i^T \beta)^2}{2\sigma^2}\right).$$

The joint conditional density is

$$p(\mathbf{Y} \mid \mathbf{X}_1, \dots, \mathbf{X}_p; \beta, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - \mathbf{x}_i^T \beta)^2}{2\sigma^2}\right).$$

Log-likelihood

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mathbf{x}_i^T \beta)^2.$$

Maximize over β

Term in $\ell(\beta, \sigma^2)$ that depends on β is

$$-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mathbf{x}_i^T \beta)^2.$$

maximizing ℓ with respect to β is equal to minimizing

$$Q(\beta) = \sum_{i=1}^n (Y_i - \mathbf{x}_i^T \beta)^2 = (\mathbf{Y} - \mathbf{X}\beta)^T (\mathbf{Y} - \mathbf{X}\beta),$$

Take the gradient of $Q(\beta)$ with respect to β and set it to zero:

$$\nabla_{\beta} Q(\beta) = -2\mathbf{X}^T (\mathbf{Y} - \mathbf{X}\beta) = 0 \Rightarrow \mathbf{X}^T \mathbf{X} \beta = \mathbf{X}^T \mathbf{Y}.$$

Considering $\mathbf{X}^T \mathbf{X}$ is invertible, the unique solution is

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}.$$

This is identical to the OLS estimator.

The Hessian of $Q(\beta)$:

$$\nabla_{\beta}^2 Q(\beta) = 2\mathbf{X}^T \mathbf{X},$$

(positive semi-definite).

Hence $Q(\beta)$ is convex, so $\hat{\beta}$ minimizes $Q(\beta)$, also maximizes $\ell(\beta, \sigma^2)$ over β .

Then we get:

$$\hat{\beta}_{MLE} = \hat{\beta}_{OLS} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}.$$

MLE is also BLUE

4 Question4

Ridge Regression

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \epsilon_i$$

$\mathbf{Y} = \mathbf{X}\beta + \epsilon$, with $\epsilon \sim N(\mathbf{0}, \sigma^2 I_n)$.

$\beta_i \sim N(0, \sigma^2/\lambda)$ for $i = 1, \dots, p$:

$$\pi(\beta) \propto \exp\left(-\frac{\lambda}{2\sigma^2} \sum_{i=1}^p \beta_i^2\right).$$

Likelihood

$$L(\beta) \propto \exp\left(-\frac{1}{2\sigma^2} \sum_{j=1}^n (Y_j - \mathbf{x}_j^T \beta)^2\right).$$

By Bayes' rule, $\pi(\beta | \mathbf{Y}) \propto L(\beta)\pi(\beta)$, so

$$\ell(\beta) = -\frac{1}{2\sigma^2} [(\mathbf{Y} - \mathbf{X}\beta)^T (\mathbf{Y} - \mathbf{X}\beta) + \lambda \beta^T D \beta] + C,$$

Also, maximizing $\ell(\beta)$ is equal to minimizing

$$Q(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^T (\mathbf{Y} - \mathbf{X}\beta) + \lambda \beta^T D \beta.$$

Setting the gradient to zero,

$$\nabla_{\beta} Q(\beta) = -2\mathbf{X}^T (\mathbf{Y} - \mathbf{X}\beta) + 2\lambda D \beta = 0 \Rightarrow (\mathbf{X}^T \mathbf{X} + \lambda D) \beta = \mathbf{X}^T \mathbf{Y},$$

giving

$$\hat{\beta}_{MAP} = (\mathbf{X}^T \mathbf{X} + \lambda D)^{-1} \mathbf{X}^T \mathbf{Y}.$$

The Hessian $\nabla_{\beta}^2 Q(\beta) = 2(\mathbf{X}^T \mathbf{X} + \lambda D)$ is positive definite for $\lambda > 0$.

So $Q(\beta)$ is strictly convex and $\hat{\beta}_{MAP}$ is the unique global maximizer.

Interpretation:

Ridge regression is just OLS with a Gaussian prior. It uses λ to shrink coefficients toward zero, trading a little accuracy for a lot of stability. This stops the model from blowing up and makes it solvable even when you have too many features.

5 Question5

LASSO Regression

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \epsilon_i$$

$\mathbf{Y} = \mathbf{X}\beta + \epsilon$, with $\epsilon \sim N(\mathbf{0}, \sigma^2 I_n)$.

$\beta_i \sim \text{Lap}(0, \sigma^2/\lambda)$ for $i = 1, \dots, p$

$$\pi(\beta_i) = \frac{\lambda}{2\sigma^2} \exp\left(-\frac{\lambda|\beta_i|}{\sigma^2}\right).$$

$$\pi(\beta) \propto \exp\left(-\frac{\lambda}{\sigma^2} \sum_{i=1}^p |\beta_i|\right).$$

Likelihood

$$L(\beta) \propto \exp\left(-\frac{1}{2\sigma^2} \sum_{j=1}^n (Y_j - \mathbf{x}_j^T \beta)^2\right).$$

By Bayes' rule, $\pi(\beta | \mathbf{Y}) \propto L(\beta)\pi(\beta)$, so

$$\ell(\beta) = -\frac{1}{2\sigma^2} (\mathbf{Y} - \mathbf{X}\beta)^T (\mathbf{Y} - \mathbf{X}\beta) - \frac{\lambda}{\sigma^2} \sum_{i=1}^p |\beta_i| + C.$$

Maximizing $\ell(\beta)$ over β is equal to minimizing

$$Q(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^T (\mathbf{Y} - \mathbf{X}\beta) + 2\lambda \sum_{i=1}^p |\beta_i| = \text{RSS}(\beta) + 2\lambda \|\beta_{1:p}\|_1.$$

LASSO objective function

Because $\hat{\beta}_{MAP}$ cannot be solved by a simple gradient equation, the maximizer satisfies the KKT condition.

$$-2\mathbf{X}^T (\mathbf{Y} - \mathbf{X}\beta) + 2\lambda \mathbf{s} = \mathbf{0},$$

Equivalently,

$$\mathbf{x}_i^T (\mathbf{Y} - \mathbf{X}\beta) = \lambda \text{sign}(\beta_i) \quad \text{if } \beta_i \neq 0, \quad |\mathbf{x}_i^T (\mathbf{Y} - \mathbf{X}\beta)| \leq \lambda \quad \text{if } \beta_i = 0.$$

$\text{RSS}(\beta)$ is convex, and $\|\beta_{1:p}\|_1$ is convex. The sum of convex functions is convex, so $Q(\beta)$ is convex. Any point satisfying the KKT condition above is a global minimizer of Q , equally a global maximizer of the posterior $\ell(\beta)$.

Interpretation:

LASSO represents the MAP estimator under a Laplace prior on β .

While the smooth Gaussian prior of Ridge regression merely shrinks all coefficients toward zero, the sharp peak of the Laplace density at zero enforces sparsity by driving irrelevant coefficients to exactly zero.

As λ grows, features drop out one by one.

6 Question6

(a)

Let $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$. Then

$$\mathbf{X}^T\mathbf{X} = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^T, \quad \mathbf{X}^T\mathbf{X} + \lambda\mathbf{I} = \mathbf{V}(\mathbf{\Sigma}^2 + \lambda\mathbf{I})\mathbf{V}^T.$$

$\mathbf{V}$ is orthogonal,

$$(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1} = \mathbf{V}(\mathbf{\Sigma}^2 + \lambda\mathbf{I})^{-1}\mathbf{V}^T.$$

So

$$\hat{\beta}^{Ridge} = \mathbf{V}(\mathbf{\Sigma}^2 + \lambda\mathbf{I})^{-1}\mathbf{V}^T\mathbf{V}\mathbf{\Sigma}\mathbf{U}^T\mathbf{y} = \mathbf{V}(\mathbf{\Sigma}^2 + \lambda\mathbf{I})^{-1}\mathbf{\Sigma}\mathbf{U}^T\mathbf{y}.$$

Therefore

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta}^{Ridge} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\mathbf{V}(\mathbf{\Sigma}^2 + \lambda\mathbf{I})^{-1}\mathbf{\Sigma}\mathbf{U}^T\mathbf{y} = \mathbf{U}\mathbf{\Sigma}(\mathbf{\Sigma}^2 + \lambda\mathbf{I})^{-1}\mathbf{\Sigma}\mathbf{U}^T\mathbf{y}.$$

Since $\mathbf{\Sigma}(\mathbf{\Sigma}^2 + \lambda\mathbf{I})^{-1}\mathbf{\Sigma} = \text{diag}(\sigma_j^2/(\sigma_j^2 + \lambda))$,

$$\hat{\mathbf{y}} = \sum_{j=1}^p \mathbf{u}_j \frac{\sigma_j^2}{\sigma_j^2 + \lambda} \mathbf{u}_j^T \mathbf{y}.$$

$\sigma_j^2/(\sigma_j^2 + \lambda)$ increases with σ_j : for small σ_j it is close to 0, for large σ_j close to 1.

So directions with smaller singular values are shrunk more, for any fixed $\lambda \geq 0$.

(b)

The ridge hat matrix: $\mathbf{H}_\lambda = \mathbf{X}(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T$.

$$\mathbf{H}_\lambda = \mathbf{U} \text{diag}\left(\frac{\sigma_j^2}{\sigma_j^2 + \lambda}\right) \mathbf{U}^T.$$

Using $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB})$ and $\mathbf{U}^T\mathbf{U} = \mathbf{I}_p$,

$$\text{tr}[\mathbf{X}(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T] = \text{tr}\left[\mathbf{U}^T\mathbf{U} \text{diag}\left(\frac{\sigma_j^2}{\sigma_j^2 + \lambda}\right)\right] = \sum_{j=1}^p \frac{\sigma_j^2}{\sigma_j^2 + \lambda}.$$

$$\text{tr}[\mathbf{X}(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T] = \sum_{j=1}^p \frac{\sigma_j^2}{\sigma_j^2 + \lambda}.$$

When $\lambda = 0$, each term equals 1 and the sum equals p .

For $\lambda > 0$, each term lies in $(0, 1)$, so the sum is strictly less than p and decreases monotonically to 0 as $\lambda \rightarrow \infty$.

7 Question7

7.1 (a)

Download the AReM data.

7.2 (b)

Train / Test Split.

7.3 (ci)time-domain

Including min, max, mean, median, std, Q1, Q3 (7 features * 6 time series = 42 features per instance)

7.4 (cii)feature extraction

Something wrong existed in the cycling/dataset9.csv and cycling/dataset14.csv.

Error tokenizing data. C error: Expected 7 fields in line 485, saw 8

File: /content/drive/MyDrive/USC_Summer26/EE559/Data/AReM/cycling/dataset14.csv

Error tokenizing data. C error: Expected 7 fields in line 485, saw 8

File: /content/drive/MyDrive/USC_Summer26/EE559/Data/AReM/cycling/dataset9.csv

Total bad files: 2

Results:

Time Series 1: avg_rss12							
Instance	min	max	mean	median	std	Q1	Q3
0	35.00	47.40	43.9545	44.33	1.5588	43.00	45.00
1	33.00	47.75	42.1798	43.50	3.6707	39.15	45.00
2	33.00	45.75	41.6781	41.75	2.2435	41.33	42.75
3	37.00	48.00	43.4550	43.25	1.3861	42.50	45.00
4	36.25	48.00	43.9691	44.50	1.6184	43.31	44.67

Time Series 2: var_rss12							
Instance	min	max	mean	median	std	Q1	Q3
0	0.0	1.70	0.4263	0.47	0.3387	0.00	0.50
1	0.0	3.00	0.6960	0.50	0.6309	0.00	1.12
2	0.0	2.83	0.5360	0.50	0.4055	0.43	0.71
3	0.0	1.58	0.3781	0.47	0.3156	0.00	0.50
4	0.0	1.50	0.4131	0.47	0.2631	0.43	0.50

Time Series 3: avg_rss13

Instance	min	max	mean	median	std	Q1	Q3
0	6.50	29.75	22.1224	23.000	3.0309	19.75	24.0000
1	8.50	30.00	22.1836	23.000	3.8105	20.50	24.3725
2	3.00	28.25	19.0066	19.125	4.0871	16.50	22.0625
3	5.75	27.00	15.7933	15.000	3.8476	13.00	18.2700
4	1.50	26.33	15.8680	16.250	3.7424	14.25	18.0000

Time Series 4: var_rss13

Instance	min	max	mean	median	std	Q1	Q3
0	0.0	4.44	0.4973	0.43	0.5507	0.00	0.83
1	0.0	5.15	0.9899	0.83	0.9537	0.43	1.30
2	0.0	6.42	0.8419	0.50	0.9288	0.43	1.12
3	0.0	10.03	0.8494	0.50	0.9958	0.43	1.12
4	0.0	5.17	0.6664	0.47	0.7890	0.00	0.94

Time Series 5: avg_rss23

Instance	min	max	mean	median	std	Q1	Q3
0	29.00	38.50	35.5885	36.00	1.9996	35.3625	36.50
1	20.00	38.67	33.4939	35.00	3.8494	30.4575	36.33
2	23.67	37.50	29.8571	30.00	2.4110	28.4575	31.25
3	8.00	33.50	23.0348	23.50	2.4889	22.2500	24.00
4	11.33	30.75	22.1038	21.67	3.3183	20.5000	23.75

Time Series 6: var_rss23

Instance	min	max	mean	median	std	Q1	Q3	label
0	0.0	1.79	0.4933	0.43	0.5135	0.00	0.94	bending1
1	0.0	2.18	0.6135	0.50	0.5243	0.00	1.00	bending1
2	0.0	1.79	0.3833	0.43	0.3892	0.00	0.50	bending1
3	0.0	5.26	0.6796	0.50	0.6225	0.43	0.87	bending1
4	0.0	2.96	0.5553	0.49	0.4878	0.00	0.83	bending1

5 rows x 43 columns

7.5 (ciii)Bootstrap Confidence

Results shown in the table below:

Feature	Std. Dev.	CI Lower (5%)	CI Upper (95%)
min_avg_rss12	8.8564	7.5743	10.1311
max_avg_rss12	4.1920	3.0131	5.1613
mean_avg_rss12	4.8254	4.2452	5.3171
median_avg_rss12	4.8730	4.2282	5.4061
std_avg_rss12	1.7571	1.5399	1.9315
q1_avg_rss12	5.6901	5.1159	6.1642
q3_avg_rss12	4.6223	3.7824	5.3478
min_var_rss12	0.0000	0.0000	0.0000
max_var_rss12	5.1442	4.6426	5.5194
mean_var_rss12	1.6052	1.4047	1.7425
median_var_rss12	1.4383	1.2442	1.5764
std_var_rss12	0.9050	0.8154	0.9656
q1_var_rss12	0.9545	0.8263	1.0481
q3_var_rss12	2.1655	1.9065	2.3396
min_avg_rss13	3.0523	2.8254	3.2216
max_avg_rss13	4.6877	3.9136	5.3259
mean_avg_rss13	3.8213	3.1458	4.3752
median_avg_rss13	3.8094	3.1158	4.3806
std_avg_rss13	1.0031	0.7880	1.1960
q1_avg_rss13	4.0999	3.4174	4.6510
q3_avg_rss13	3.9196	3.2164	4.5032
min_var_rss13	0.0000	0.0000	0.0000
max_var_rss13	2.3023	2.0655	2.4877
mean_var_rss13	1.1827	1.0823	1.2398
median_var_rss13	1.1534	1.0517	1.2087
std_var_rss13	0.4743	0.4313	0.5057
q1_var_rss13	0.8442	0.7660	0.8956
q3_var_rss13	1.5701	1.4354	1.6475
min_avg_rss23	5.3433	3.3536	6.9368
max_avg_rss23	5.4893	4.2958	6.4153
mean_avg_rss23	5.1498	3.6282	6.3434
median_avg_rss23	5.2998	3.6831	6.5568
std_avg_rss23	1.0424	0.7775	1.2652
q1_avg_rss23	5.5681	4.0234	6.7848
q3_avg_rss23	4.9930	3.5932	6.1568
min_var_rss23	0.0521	0.0000	0.0890
max_var_rss23	2.5590	2.2430	2.8452
mean_var_rss23	1.1743	1.0700	1.2312
median_var_rss23	1.1088	1.0062	1.1668
std_var_rss23	0.5186	0.4761	0.5463
q1_var_rss23	0.7780	9 0.7035	0.8261
q3_var_rss23	1.5545	1.4197	1.6270

7.6 (civ)Selection

Feature	Separation Score
max_avg_rss23	2.4866
mean_avg_rss23	1.9848
q3_avg_rss23	1.9647
median_avg_rss23	1.8711
q1_avg_rss23	1.8163
min_avg_rss23	1.7494

Six of the top six features all come from the same time series, **avg_rss23**, strongly s is the most discriminative for distinguishing bending from other activities.

Then, among the six **avg_rss23** features, **min**, **mean**, and **max** were chosen.

7.7 (cv)Plots

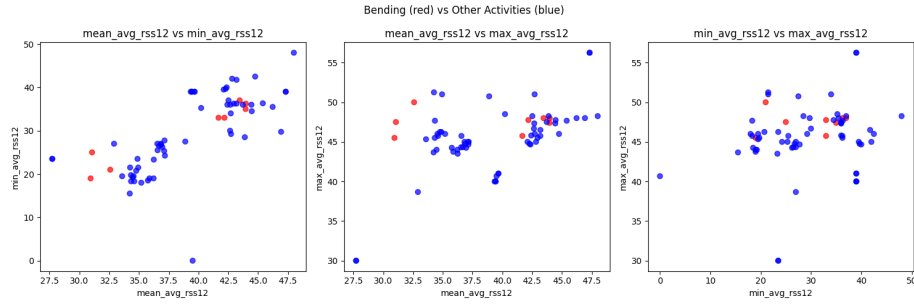


Figure 1: Scatter plots of **mean_avg_rss12**, **min_avg_rss12**, and **max_avg_rss12** for the training instances, comparing bending activities (red) against all other activities (blue).